

4G LTE

Module 43

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hello viewers I welcome you to the series of lectures on cyber security and information security today we will discuss 4G LTE LTE stands for long-term Evolution and it more applies for improving Wireless Broadband speeds to meet increasing demand LTE is 4G Wireless Communications standard developed by the third generation partnership project 3gpp 3gpp Engineers named the technology long-term Evolution because it represents the next step in progression for GSM a 2G standard to UMTS that is universal mobile telecommunications service the 3G Technologies based upon GSM the goal of LTE was to increase the capacity and speed of wireless data networks using new DSP that is digital signal processing techniques and modulations it is designed to provide up

to 10x the speeds of 3G networks for mobile devices such as smartphones tablets notebooks and wireless hotspots so now what is the goal a goal was the redesign and simplification of the network architecture to an IP based system with significantly reduced transfer latency compared to the 3G architecture evolution of 4G LTE tracking the evolution of Wireless technology with the evolution we get changes to the existing system as in mobile network lot of evolution is occurring in 1973 the first call was made by engineer Martin Cooper from their Motorola Mobile phone after that in 1987 actor Michael Douglas in the movie Wall Street playing a Financial Titan talking over his mobile phone from the beach the first smartphone Simon was

introduced to the world in 1992 as the first wireless device to the Imagine functions Beyond making phone calls Simon's Innovation was ahead of its time but it is not ready to handle Simon's capabilities after the next two decades a series of mobile network technology standards came to Market the first generation of mobile Communications started with the advanced mobile phone systems a Ms it is a 1G from there we move to GSM and CDMA 1 and then to UMTS and EV Doo these are 3G Technologies the latest Technologies of 4G are LTE from the 3G PP group and 802.16m from i e in the case of 802.16m the candidate for 4G is also known as Wireless MN Advanced or ymax 2 LTE progresses through versions known as releases the latest release that

qualifies as being 4G is release 10 often called as LTE advanced 4 minutes, 1 second 4G LTE enables Advanced Technologies introduced from the last three years in transportation healthcare small business Enterprise and education Solutions like intuitive go payment enable consumers and businesses to experience the power of 4G LTE through go payments small businesses such as Street vendors farmer markets and food trucks can attach a small card reader to a smartphone or tablet making payment transactions quick and easy for the vendor and customer with the last 40 Years of mobile technology in the rear view mirror and the large scale deployment of V LTE and LTE Advanced on the horizon one can only imagine what the next 40 years will hold LTE technical

objectives user through port in megahertz down link Uplink down link capacity Uplink capacity latency that is transition time less than 5 millisecond in ideal conditions that is user plan 100 millisecond control plane fast connection setup simplified architecture simpler e u t r n architecture no RNC C no CS domain no DCH Mobility optimized for low speed but supporting 120 km/ hour now what is the difference between 4G and LTE 4G LTE has several competing 4G standards with ultra mobile broadband UMB and yaax i e 802.16 4G Technologies are designed to provide IP based voice data and multimedia streaming at speeds of at least megabit per second and up to as fast as 1 gbit per second the leading cell providers has started to deploy 4G

Technologies with verizone and AT and T launching 4G LTE networks and Sprint utilizing its new 4G ymax network in terms of mobile devices most of the newer Android based smartphones are 4G LTE capable the iPhone 5 and iPad 3 are expected to have built-in 4G LTE capabilities when released in the second half of 2012 speed the difference between slow 3D networks and new 4G or LTE networks is faster most of the 4G and true 4G networks have upload and download speeds that are almost identical but now LT is the fastest connection available for wireless networks resources must for 4G LTE technology for creating 4G connectivity it requires two components first Network that can support the necessary speeds

second device that is able able to connect to that Network and download information at a high enough speed but just because a phone has 4G LTE connectivity inside doesn't mean you can get the speeds you want in the same way that buying a car that can drive 200 m per hour doesn't mean you can go fast on 55 m per our freeway is LT is same as 4G or not no LT is not same as 4G there is 4G and then there is 4G LTE 4G stands for fourth generation which comes next to the previous 3G networks it is much faster and with a greater capacity 4G LT stands for fourth generation long-term Evolution which is the purest form of 4G technology as if a cellular provider describes a 4G network without mentioning LTE they are probably talking about a high speed packet excess

Network the HSPA network is a faster version of 3G gsm network while not as fast as LTE network it is still faster than a 3G network advantages of Lte provides low latency High Network throughput increased data transfer speed more cost Effectiveness improvements

over 3G network devices which are covered under LTE or mobile phones laptops cameras camcorders assured interoperability with older Wireless Technologies such as GSM wcdma HSPA CDMA TDS CDMA LTE will ensure the success of mobile internet difference between different Generations 4G stands for fourth generation of Cellular Communications 10 times faster than current 3G so one don't need Broadband in fact many things could be done through smartphones the main reason 4G is faster than 3G is because of orthogon frequency

division multiplexing that is ofdm ofdm is a technique for squeezing more data onto the same amount of radio frequency it also reduces latency and interference 4G networks use different frequencies to transmit data than 3G so you need a handset which has a modem that supports these new frequencies LTE is a 4G Wireless Broadband technology developed by the third generation partnership project 3GPP LTE provides significantly increased Peak data rates with the potential for 100 megabits per second Downstream and 30 megabits per second Upstream reduced latency scalable bandwidth capacity and backwards compatibility with existing GSM and UMTS technology VTE it enables Wireless carriers to use IP based 4G LTE data networks for voice calls like WhatsApp call video call

conferencing Etc 5G fifth generation of cellular communication 5G networks are aimed at meeting the requirements for mobile Communications Beyond 2020 the speed of 5G is much faster than 4G 20 GB per second fast enough to download HD movies in a span of seconds so all of the above are expected to be driver for Trends such as the internet of things and initiatives such as smart cities and digital India and revolutionize India a speedy way to reach our 2020 vision of apj clam disadvantages of 4G LTE it has advanced data rates for customers customers are required to purchase a new equipment to maintain the 4G the 4G is only presently accessible in convinced cities needs complex Hardware high definition voice the most noticeable benefit consumers will experience with

VoLTE is improved voice quality traditional voice networks transmit voice calls using a 8 kilobits per second codec Varone says it's using a 13 KB PS codec that also uses more modern compression methods Rich communication services or RCS VTE also will enable wireless operators to deliver a new set of Standards Based Services refer to as Rich Communications services or RCS these Services include things like video calling file transferring realtime language translation video voicemail and instant messaging many of these applications are already delivered over cellular data service using the third party apps for instance iPhone users are very familiar with FaceTime for video calling millions of people use Skype for video calls as well as messenger services and Google offers it Google

translate service the benefit of VTE is that it lets you launch these Services directly from the phone's native dialer faster call setup times connecting calls using VoLTE will be much faster than it is today using the older circuit switch technology on a 2G wireless network improved battery life over other VoIP apps a report from signals research group suggests that VTE services offer much improved battery life compared with using an over the-top voice over IP

service such as Skype VTE delivered a consistently higher call Quality than circuit switched voice and over the top applications in addition VTE really shined when it came to its Network requirements consuming far less than a Skype voice call which translates into a meaningfully longer smartphone battery

life integration of VTE with voice over Wi-Fi service another benefit for consumers is better integration between a wireless operators cellular voice service and a Wi-Fi calling service this is particularly important for operators which are starting to use Wi-Fi to handle some voice calls as well as other data traffic improving the transition between the cellular voice call which uses licensed Wireless spectrum and a Wi-Fi call which uses unlicensed spectrum is important because it could let these operators get better indoor coverage in areas where their licensed Wireless spectrum is limited that means improved coverage for consumers because the call setup necessary for voice over Wi-Fi is the same as the setup for

VTE once volte is fully deployed it will be easier and more seamless to offer voice over Wi-Fi in areas where licensed spectrum is either weak or unavailable true device interoperability once carriers make VT Services interoperable among different carrier networks there won't be any technology barriers among most of the world's Wireless operators the CDMA GSM divide that separated the four major Wireless operators in the US will no longer be an issue what this could mean for the average consumer is that you may be able to buy one device that could truly operate on any wireless network all the major carriers would be delivering data services and voice services using the same 4G LT technology and if your smartphone supported enough LT Wireless

bands you would be able to take that device to any carrier and get both voice and data service service now security of 17 minutes, 1 second 4G LTE LTE security it is a term that provides a powerful defense mechanism versus possible threats from the internet imposed by multiple types of attacks confidentiality means making sure that the right people can in fact get the right data integrity means the message sent from party a to the party B did not change these are the essential points to highlight the security issues of LTE technology for the Next Generation in Mobile Communication in order to provide more powerful security and that will be done through achieving the sub objectives to discover the main factors that used to compare the LTE cryptographic

algorithms to find out these trends and weaknesses of each LTE cryptographic algorithm to evaluate which algorithm provide higher security than others to propose an algorithm from the existing LTE cryptographic algorithms and that may enhance the security of the existing LD cryptographic algorithms so that's all for 4G LTE thank you viewers for watching this program